

BTH545 Workshop 1

Learning Outcomes

- Install Qt Creator
- Understand the steps in building and compiling a project with Qt creator
- Create an execute a simple Qt GUI application.

Lab (60 %)

In this workshop, you have two goals:

1. To download and install Qt Creator
2. To create a simple Qt program that displays a button the text “Hello BTH545”

Select the open-source download page and then go to the binary section. Select the installer for your operating system. You should then download and run the installer for your operating system. It will lead you through the installation process.

Once you have Qt Creator installed, you should write a program to create a button that displays the text “Hello BTH545”.

Reflection (40%)

Answer the following questions in a text file called reflect.txt and submit it along with the workshop. Your answers should be significant. The reflections should explore the topics in depth and not just give single line answers.

1. When you create the push button explain how the parameters for the constructor determine that the button will appear on the desktop rather than as a child of another widget.
2. Explain what happens when you call the .show() method on the push button and give your opinion as to why the API was designed to require a show method.
3. You show the push button yet you call exec on the application. This seems strange. Explain what you think the relationship between the application and the pushbutton must be for this to work the way it does.

Submission

Some parts may be different in your system. Once you have the program written, you should deploy it with the required libraries so that it can be run on any windows system. Your Qt installation comes with a program called windeployqt which should be in the bin directory for the compiler you are using. On typical installations, it is located in:

C:\Qt\5.11.0\mingw53_32\bin

Do note that the above version number will change. Once you locate the bin directory of your installation, you should follow the steps below.

1. In the project tab for your project, set the build type to be release



Build Settings

Edit build configuration:

Release ▼

Add ▼

Remove

Rename...

Clone...

2. Then, click build and wait for it to finish.

3. Locate the build directory for the release version of the project and navigate to the sub directory for the compiler version you are using and then to the release directory.
4. Copy all .exe and .o files to a new, empty directory.
5. Start a command prompt and navigate to the bin dir for your Qt installation. Execute the program qtenv2.bat
6. Now execute windeployqt <bin_dir>, where bin_dir is the directory where you placed the .exe and .o files for your project.
7. Use a file browser to inspect the bin_dir and you should see .dll files have been added.
8. Double click on the .exe file and your program should run.
9. Zip the entire bin_dir so that it will unpack with the directory name at the top and the files and directories within it.
10. Attach the resulting zip file to your assignment on Blackboard for submission. Also, attach the reflect.txt file to the Blackboard submission page.